UNIVERSITY OF ENGINEERING AND TECHNOLOGY, LAHORE.

Course: Psychology (HU-214) Course Instructor: Sehar Farooq

Week: 3-4
Topic: Thinking and Intelligence

a) Intelligence - is there just one type?

In many studies, intelligence has been defined as whatever intelligence tests measure, which has tended to be school smarts. But intelligence is not a quality like height or weight, which has the same meaning to everyone worldwide. People assign the term intelligence to the qualities that enable success in their own time and culture.

Howard Gardner has identified **eight** relatively independent intelligences, including the verbal and mathematical aptitudes assessed by standard tests.

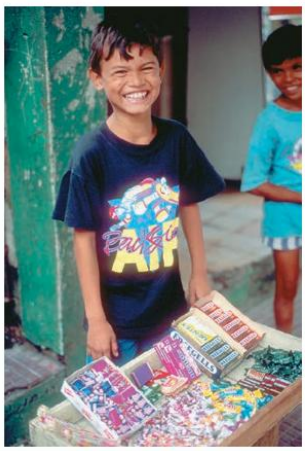


Gardner has also proposed a ninth possible intelligence—existential intelligence—the ability to ponder deep questions about life.

Sternberg's Three Intelligences: Robert Sternberg (1985, 2011) agrees with Gardner that success involves more than traditional intelligence and that we have multiple intelligences. But his triarchic theory proposes three intelligences:

1. ***Analytical*** (academic problem - solving) intelligence is assessed by intelligence tests, which present well - defined problems having a single right answer. Such tests predict school grades reasonably well and vocational success to a lesser extent.

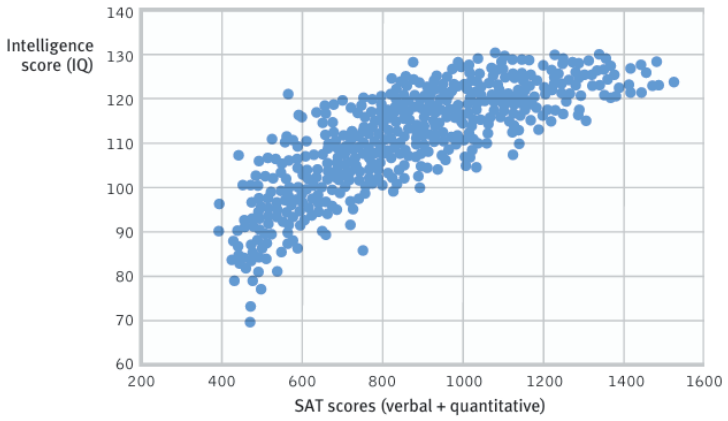
2. ***Creative*** intelligence is demonstrated in innovative smarts: the ability to generate novel ideas. "You're wise, but you lack tree smarts."
3. ***Practical*** intelligence is required for everyday tasks that are not well-defined and that may have many possible solutions. Managerial success, for example, depends less on academic problem-solving skills than on a shrewd ability to manage oneself, one's tasks, and other people.



Street smarts This child selling candy on the streets of Manaus, Brazil, is developing practical intelligence at a very young age.

b) Intelligence: How do we best measure it?

An intelligence test assesses people's mental abilities and compares them with others, using numerical scores. How do we design such tests, and what makes them credible? Consider why psychologists created tests of mental abilities and how they have used them. By this point in your life, you've faced dozens of ability tests: school tests of basic reading and math skills, course exams, intelligence tests, driver's license exams.



Psychologists classify such tests as either ***achievement tests,*** intended to reflect what you have learned, or ***aptitude tests,*** intended to predict your ability to learn a new skill. Exams covering what you have learned in this course are achievement tests. A college entrance exam, which seeks to predict your ability to do college work, is an aptitude test—a *"thinly disguised intelligence test."*

c) Measures of Intelligence

Intelligence Tests developed over time and their modern-day usage.

Era	Test	Developer(s)	Main Contribution
1905	Binet-Simon Scale	Alfred Binet & Théodore Simon	First practical intelligence test; identified students needing educational support.
1916	Stanford-Binet Intelligence Scale	Lewis Terman	Adapted Binet's test for the U.S.; introduced the concept of IQ.
1917–1918	Army Alpha & Beta Tests	Robert Yerkes	First large-scale group intelligence testing.
1939	Wechsler-Bellevue Scale	David Wechsler	Broadened intelligence assessment beyond academic skills.

1949	Wechsler Intelligence Scale for Children (WISC)	David Wechsler	Specialized IQ test for children.
1955	Wechsler Adult Intelligence Scale (WAIS)	David Wechsler	Became the leading adult intelligence assessment.
1938 onward	Raven's Progressive Matrices	John C. Raven	Measures abstract reasoning; minimizes language and cultural bias.
1967	Cattell Culture Fair Intelligence Test	Raymond Cattell	Designed to reduce cultural influence.
2000s-Present	Modern revisions (WAIS-IV/V, WISC-V, Stanford-Binet 5)	Various	More comprehensive measurement of cognitive abilities.

Mental age *is a measure of intelligence test performance devised by Binet; the chronological age that most typically corresponds to a given level of performance. Thus, a child who does as well as an average 8-year-old is said to have a mental age of 8.*

d) Heritability and intelligence

Heritability and intelligence refer to the extent to which genetic differences among individuals contribute to variations in intelligence within a population. Research from twin, adoption, and family studies suggests that ***intelligence is moderately to highly heritable,*** with genetic factors accounting for a significant proportion of the differences in IQ scores. This does not mean intelligence is determined solely by genes; heredity provides a potential range of intellectual ability, while environmental factors such as education, nutrition, socioeconomic conditions, and life experiences shape how that potential is expressed. Therefore, intelligence is best understood as the result of a complex interaction between genetic inheritance and environmental influences, making heritability and intelligence positively correlated but not exclusively linked.

e) Problem solving, decision making, reasoning

There are several factors that can simultaneously affect a person's intellect and ability to solve problems. Some of them are discussed below:

Age

- **Fluid intelligence** (problem-solving, processing speed, abstract reasoning) tends to peak in early adulthood and gradually decline with age.
- **Crystallized intelligence** (knowledge, vocabulary, wisdom) often continues to increase into later adulthood.

Gender

- Overall intelligence levels are generally similar between males and females.
- Small average differences may appear in specific cognitive abilities (e.g., some verbal or spatial tasks), but these do not indicate differences in overall intelligence.

The most reliable male edge appears in spatial ability tests like Tetris, the Rubik's Cube, and so on. The solution requires speedily rotating three-dimensional objects in one's mind. Today, such skills help when fitting suitcases into a car trunk, playing chess, or doing certain types of geometry problems. From an evolutionary perspective, those same skills would have helped our ancestral fathers track prey and make their way home. The survival of our ancestral mothers may have benefited more from a keen memory for the location of edible plants—a

legacy that lives today in women's superior memory for objects and their location. But experience matters.

Race and Ethnicity

- Modern psychology recognizes that intelligence is influenced by a complex interaction of genetic, environmental, cultural, educational, and socioeconomic factors.
- Differences observed in test performance are often linked to factors such as educational opportunities, socioeconomic conditions, health, and cultural experiences rather than race itself.

f) Emotional and Social Intelligence

Emotional Intelligence is the ability to perceive, understand, manage, and use emotions effectively in oneself and others. Popularized by Daniel Goleman, it includes self-awareness, self-regulation, motivation, empathy, and social skills.

Individuals with high emotional intelligence are better *able to manage stress, build positive relationships, communicate effectively, and make sound decisions.* Consequently, emotional intelligence plays an important role in personal, social, and professional success.

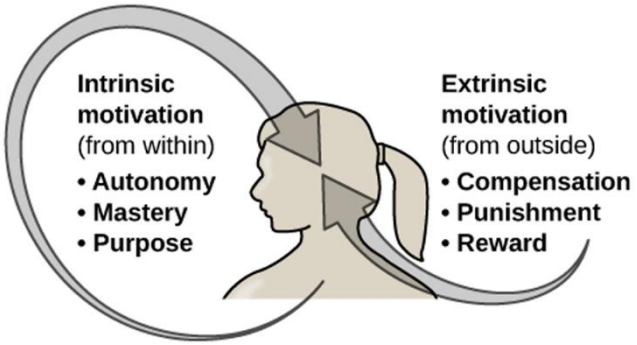
Social Intelligence is the ability to understand and effectively navigate social situations and relationships. Introduced by Edward Thorndike, it involves social awareness, empathy, interpersonal skills, relationship management, and adaptability.

Individuals with high social intelligence *can interpret social cues, communicate effectively, build strong relationships, and work successfully with others.* Social intelligence is essential for leadership, teamwork, and social adjustment.

Topic: Emotion and Motivation

a) Cultivating motivation

Motivation describes the wants or needs that direct behavior toward a goal. In addition to biological motives, motivations can be ***intrinsic*** (arising from internal factors) or ***extrinsic*** (arising from external factors). Intrinsically motivated behaviors are performed because of the sense of personal satisfaction that they bring, while extrinsically motivated behaviors are performed in order to receive something from others.



In reality, our motivations are often a mix of both intrinsic and extrinsic factors, but the nature of the mix of these factors might change over time.

Instinct Theory of Motivation (William James)

William James (1842–1910) was an important contributor to early research into motivation, and he is often referred to as the father of psychology in the United States. According to James, people are born with natural tendencies that motivate them to act in certain ways to aid survival and adaptation. These instincts automatically direct behavior *without the need for learning.*

James believed humans possess many instincts, including curiosity, fear, sociability, parental care, and the desire for achievement. From a biological perspective, an instinct is a species-specific pattern of behavior that is not learned.

While the theory effectively explains many automatic and survival-oriented behaviors, it cannot fully explain complex human actions that are influenced by learning, culture, and personal experiences.

Drive Theory of Motivation (Clark Hull)

This theory states that behavior is motivated by the desire to reduce internal states of tension caused by unmet physiological needs. These needs create drives, such as hunger or thirst, which energize behavior directed toward restoring the body's state of balance, known as homeostasis. According to the theory, *individuals are motivated to act in ways that satisfy their biological needs and reduce discomfort.*

While Drive Theory effectively explains basic physiological motivations, it is less successful in explaining complex behaviors that are driven by curiosity, achievement, or the pursuit of excitement.

Expected Questions

Short Answers

- What are Sternberg's three types of intelligences? Explain any one.
- Is intelligence heritable?
- What are key differences between social and emotional intelligence?
- Differentiate between intrinsic and extrinsic motivation.

Long Answers

- What are the most prominent measures of intelligence developed over time?
- Explain the factors that may affect intelligence, problem solving, and decision making.
- Explain the instinct theory of motivation by William James.
- Explain the drive theory of motivation by Clark Hull.